CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Trigonometric ratios

1. If $\sin\theta = \frac{a}{b}$, write $\cos\theta$ in terms of a and b. (a) $\frac{b}{\sqrt{b^2-a^2}}$ (b) $\frac{b}{a}$ (c) $\frac{\sqrt{b^2-a^2}}{b}$ (d) $\frac{a}{\sqrt{b^2-a^2}}$

CBSE 2026, 1 mark

SOLUTION

Take a right triangle with the side opposite θ equal to a and the hypotenuse equal to b.

The adjacent side is $\sqrt{b^2-a^2}$, so $\cos\theta = \frac{\sqrt{b^2-a^2}}{b}$.

Answer: (c)

2. If $\tan\theta = \frac{24}{7}$, find $\sin\theta + \cos\theta$. CBSE 2026, 2 marks

SOLUTION

Take the opposite side as 24 and the adjacent side as 7. Hypotenuse $= \sqrt{24^2+7^2} = \sqrt{625} = 25$.

$\sin\theta = \frac{24}{25}$ and $\cos\theta = \frac{7}{25}$

Answer: $\frac{31}{25}$

3. If $\cot\theta = \frac{7}{8}$, evaluate $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$. CBSE 2026, 2 marks

SOLUTION

Numerator $= 1-\sin^2\theta = \cos^2\theta$ and denominator $= 1-\cos^2\theta = \sin^2\theta$

Value $= \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \left(\frac{7}{8}\right)^2$

Answer: $\frac{49}{64}$

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4. If $2\tan A = 3$, find the value of $\dfrac{4\sin A + 3\cos A}{4\sin A - 3\cos A}$. (a) $\dfrac{7}{\sqrt{13}}$ (b) $\dfrac{1}{\sqrt{13}}$ (c) 3 (d) does not exist

CBSE 2023, 1 mark

SOLUTION

Divide the numerator and the denominator by $\cos A$: $\dfrac{4\tan A + 3}{4\tan A - 3}$

$\tan A = \dfrac{3}{2}$, so $4\tan A = 6$ and the value is $\dfrac{6+3}{6-3} = 3$.

Answer: (c)

5. If $\sin\alpha = \dfrac{1}{\sqrt{2}}$ and $\cot\beta = \sqrt{3}$, find $\operatorname{cosec}\alpha + \operatorname{cosec}\beta$. CBSE 2023, 2 marks

SOLUTION

$\operatorname{cosec}\alpha = \dfrac{1}{\sin\alpha} = \sqrt{2}$

$\cot\beta = \sqrt{3}$ gives $\beta = 30^\circ$, so $\operatorname{cosec}\beta = 2$.

Answer: $2 + \sqrt{2}$

6. If $4\tan\theta = 3$, evaluate $\dfrac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}$. CBSE 2018, 3 marks

SOLUTION

$\tan\theta = \dfrac{3}{4}$. Take the opposite side as 3 and the adjacent side as 4, so the hypotenuse is 5.

$\sin\theta = \dfrac{3}{5}$ and $\cos\theta = \dfrac{4}{5}$

Numerator $= \dfrac{12}{5} - \dfrac{4}{5} + 1 = \dfrac{13}{5}$ and denominator $= \dfrac{12}{5} + \dfrac{4}{5} - 1 = \dfrac{11}{5}$

Answer: $\dfrac{13}{11}$

2 Ratios of standard angles

7. If $\cos A = \dfrac{1}{2}$, find the value of $\sin^2 A + 2\cos^2 A$. (a) $\dfrac{3}{2}$ (b) $\dfrac{5}{4}$ (c) -1 (d) $\dfrac{1}{2}$ CBSE 2026, 1 mark

SOLUTION

$\cos^2 A = \dfrac{1}{4}$, so $\sin^2 A = 1 - \dfrac{1}{4} = \dfrac{3}{4}$

$\sin^2 A + 2\cos^2 A = \dfrac{3}{4} + \dfrac{2}{4} = \dfrac{5}{4}$

Answer: (b)

8. Find the acute angle θ for which $7 + 4\sin\theta = 9$. CBSE 2025, 1 mark

SOLUTION

$4\sin\theta = 2$, so $\sin\theta = \dfrac{1}{2}$

Answer: $\theta = 30^\circ$

9. Given that $x\cos 60^\circ + y\cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. CBSE 2025, 2 marks

SOLUTION

Put in the values: $x \times \frac{1}{2} + y \times 1 + \frac{1}{2} - 1 = 5$

$\frac{x}{2} + y = \frac{11}{2}$. Multiply both sides by 2.

Answer: $x + 2y = 11$

10. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. CBSE 2025, 2 marks

SOLUTION

$\tan 60^\circ = \sqrt{3}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$ and $\cos 30^\circ = \frac{\sqrt{3}}{2}$

Value $= \frac{3}{\frac{3}{4} + \frac{3}{4}} = \frac{3}{\frac{3}{2}}$

Answer: 2

11. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ CBSE 2024, 1 mark

SOLUTION

$\cos(\alpha + \beta) = 0 = \cos 90^\circ$, so $\alpha + \beta = 90^\circ$.

$\cos\left(\frac{\alpha + \beta}{2}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$

Answer: (a)

12. Find the value of $2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ$. CBSE 2024, 2 marks

SOLUTION

$= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2}$

$= 1 + 3$

Answer: 4

13. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. CBSE 2024, 2 marks

SOLUTION

LHS $= \sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$

RHS $= \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$

LHS = RHS. Verified.

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14. Evaluate $\frac{3}{4}\tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ$. (a) -1 (b) $\frac{5}{6}$ (c) $-\frac{3}{2}$ (d) $\frac{1}{6}$ CBSE 2023, 1 mark

SOLUTION

$\tan^2 30^\circ = \frac{1}{3}$, $\sec^2 45^\circ = 2$ and $\sin^2 60^\circ = \frac{3}{4}$

Value $= \frac{3}{4} \times \frac{1}{3} - 2 + \frac{3}{4} = \frac{1}{4} - 2 + \frac{3}{4} = -1$

Answer: (a)

3 Trigonometric identities

15. Given $x = h + a\cos\theta$ and $y = k + b\sin\theta$, prove that $\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1$. CBSE 2026, 3 marks

SOLUTION

From the first relation, $x - h = a\cos\theta$, so $\frac{x-h}{a} = \cos\theta$.

From the second, $y - k = b\sin\theta$, so $\frac{y-k}{b} = \sin\theta$.

Squaring and adding: LHS $= \cos^2\theta + \sin^2\theta = 1$ = RHS. **Proved.**

16. Prove that $\frac{\tan A}{1+\sec A} - \frac{\tan A}{1-\sec A} = 2\,\text{cosec}\,A$. CBSE 2026, 3 marks

SOLUTION

$$\text{LHS} = \tan A\left[\frac{(1-\sec A)-(1+\sec A)}{(1+\sec A)(1-\sec A)}\right] = \frac{-2\tan A\sec A}{1-\sec^2 A}$$

Since $\sec^2 A - 1 = \tan^2 A$, the denominator is $-\tan^2 A$.

$$\text{LHS} = \frac{-2\tan A\sec A}{-\tan^2 A} = \frac{2\sec A}{\tan A} = \frac{2}{\cos A}\times\frac{\cos A}{\sin A} = \frac{2}{\sin A}$$

$= 2\,\text{cosec}\,A$ = RHS. **Proved.**

17. Simplify $\tan^2\theta - \left(\frac{1}{\cos\theta}\times\sec\theta\right)$. CBSE 2025, 1 mark

SOLUTION

$\frac{1}{\cos\theta} = \sec\theta$, so the bracket is $\sec^2\theta$.

$\tan^2\theta - \sec^2\theta = -(\sec^2\theta - \tan^2\theta) = -1$

Answer: -1

18. Prove that $\frac{\tan\theta}{1-\cot\theta}+\frac{\cot\theta}{1-\tan\theta}=1+\sec\theta\,\text{cosec}\,\theta$. CBSE 2025, 3 marks

SOLUTION

Write everything in terms of $\sin\theta$ and $\cos\theta$.

First term: $\frac{\sin\theta}{\cos\theta}\div\frac{\sin\theta-\cos\theta}{\sin\theta}=\frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)}$

Second term: $\frac{\cos\theta}{\sin\theta}\div\frac{\cos\theta-\sin\theta}{\cos\theta}=-\frac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)}$

LHS $=\frac{\sin^3\theta-\cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$

Using $a^3-b^3=(a-b)(a^2+ab+b^2)$: LHS $=\frac{\sin^2\theta+\sin\theta\cos\theta+\cos^2\theta}{\sin\theta\cos\theta}=\frac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta}$

$=\frac{1}{\sin\theta\cos\theta}+1=1+\sec\theta\,\text{cosec}\,\theta$ = RHS. **Proved.**

19. Prove that $\frac{\sin A+\cos A}{\sin A-\cos A}+\frac{\sin A-\cos A}{\sin A+\cos A}=\frac{2}{2\sin^2 A-1}$. CBSE 2025, 3 marks

SOLUTION

Take the common denominator $(\sin A-\cos A)(\sin A+\cos A)=\sin^2 A-\cos^2 A$.

Numerator $=(\sin A+\cos A)^2+(\sin A-\cos A)^2=2\sin^2 A+2\cos^2 A=2$

LHS $=\frac{2}{\sin^2 A-\cos^2 A}=\frac{2}{\sin^2 A-(1-\sin^2 A)}$

$=\frac{2}{2\sin^2 A-1}$ = RHS. **Proved.**

20. Given $\sec\theta-\tan\theta=m$, find $\sec\theta+\tan\theta$. (a) $1-\frac{1}{m}$ (b) m^2-1 (c) $\frac{1}{m}$ (d) $-m$ CBSE 2024, 1 mark

SOLUTION

$\sec^2\theta-\tan^2\theta=1$, so $(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)=1$.

$\sec\theta+\tan\theta=\frac{1}{\sec\theta-\tan\theta}=\frac{1}{m}$

Answer: (c)

21. Prove the identity $\frac{\tan\theta}{1-\cot\theta}+\frac{\cot\theta}{1-\tan\theta}=1+\sec\theta\,\text{cosec}\,\theta$. CBSE 2024, 3 marks

SOLUTION

Write $s=\sin\theta$ and $c=\cos\theta$. Then $\frac{\tan\theta}{1-\cot\theta}=\frac{s/c}{(s-c)/s}=\frac{s^2}{c(s-c)}$ and $\frac{\cot\theta}{1-\tan\theta}=\frac{c/s}{(c-s)/c}=-\frac{c^2}{s(s-c)}$.

LHS $=\frac{s^3-c^3}{sc(s-c)}=\frac{(s-c)(s^2+sc+c^2)}{sc(s-c)}=\frac{s^2+c^2+sc}{sc}$

$=\frac{1+sc}{sc}=\frac{1}{sc}+1$

$=1+\sec\theta\,\text{cosec}\,\theta=$ **RHS**

22. If $\sin\theta + \cos\theta = \sqrt{3}$, find the value of $\sin\theta\cos\theta$. CBSE 2023, 2 marks

SOLUTION

Square both sides: $\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 3$

$1 + 2\sin\theta\cos\theta = 3$

Answer: $\sin\theta\cos\theta = 1$

23. Prove that $\dfrac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = \dfrac{1 + \sin\theta}{\cos\theta}$. CBSE 2023, 3 marks

SOLUTION

In the numerator, replace 1 by $\sec^2\theta - \tan^2\theta$.

Numerator $= (\sec\theta + \tan\theta) - (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = (\sec\theta + \tan\theta)(1 - \sec\theta + \tan\theta)$

The second bracket is the same as the denominator, so LHS $= \sec\theta + \tan\theta$.

$= \dfrac{1}{\cos\theta} + \dfrac{\sin\theta}{\cos\theta} = \dfrac{1 + \sin\theta}{\cos\theta}$ = RHS. **Proved.**

24. Prove that $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. CBSE 2019, 3 marks

SOLUTION

LHS $= \sin^2\theta + \operatorname{cosec}^2\theta + 2\sin\theta\operatorname{cosec}\theta + \cos^2\theta + \sec^2\theta + 2\cos\theta\sec\theta$

$\sin\theta\operatorname{cosec}\theta = 1$ and $\cos\theta\sec\theta = 1$, so LHS $= (\sin^2\theta + \cos^2\theta) + 4 + \operatorname{cosec}^2\theta + \sec^2\theta$

$= 1 + 4 + (1 + \cot^2\theta) + (1 + \tan^2\theta)$

$= 7 + \tan^2\theta + \cot^2\theta$ = RHS. **Proved.**

25. Prove that $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. CBSE 2019, 3 marks

SOLUTION

LHS $= \left(\dfrac{\sin A + \cos A - 1}{\sin A}\right)\left(\dfrac{\cos A + \sin A + 1}{\cos A}\right)$

$= \dfrac{(\sin A + \cos A)^2 - 1}{\sin A\cos A} = \dfrac{\sin^2 A + \cos^2 A + 2\sin A\cos A - 1}{\sin A\cos A}$

$= \dfrac{2\sin A\cos A}{\sin A\cos A} = 2$ = RHS. **Proved.**

26. Prove that $\dfrac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \dfrac{1}{\sec A - \tan A}$. CBSE 2019, 4 marks

SOLUTION

Divide the numerator and the denominator by $\cos A$: LHS $= \dfrac{\tan A - 1 + \sec A}{\tan A + 1 - \sec A}$

In the denominator, write $1 = \sec^2 A - \tan^2 A = (\sec A - \tan A)(\sec A + \tan A)$.

Denominator $= (\tan A - \sec A) + (\sec A - \tan A)(\sec A + \tan A) = (\sec A - \tan A)(\sec A + \tan A - 1)$

LHS $= \dfrac{\tan A + \sec A - 1}{(\sec A - \tan A)(\sec A + \tan A - 1)} = \dfrac{1}{\sec A - \tan A}$ = RHS. **Proved.**

27. Prove that $\dfrac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$. CBSE 2018, 4 marks

SOLUTION

LHS $= \dfrac{\sin A(1 - 2\sin^2 A)}{\cos A(2\cos^2 A - 1)}$

Using $1 = \sin^2 A + \cos^2 A$: $1 - 2\sin^2 A = \cos^2 A - \sin^2 A$ and $2\cos^2 A - 1 = \cos^2 A - \sin^2 A$.

The two brackets are equal and cancel: LHS $= \dfrac{\sin A}{\cos A} = \tan A$ = RHS. **Proved.**

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